Unit - IV

Dynamic Programming: General method, **Applications -** Matrix chain multiplication, Optimal binary search trees, 0/1 knapsack problem, All pairs shortest path problem, Travelling sales person problem.

Dynamic Programming

Principle of optimality :

Definition: A problem is said to satisfy the Principle of Optimality if **the subsolutions of an optimal solution of the problem are themesleves optimal solutions for their subproblems**.

1.Matrix Chain Multiplication

- Matrix-chain multiplication problem
  - Given a chain $A_1, A_2, \ldots, A_n$ of n matrices, where for $i=1, 2, \ldots, n$, matrix A_i has dimension $p_{i-1} \times p_i$
  - Parenthesize the product $A_1 A_2 \ldots A_n$ such that the total number of scalar multiplications is minimized.

Matrix Multiplication

$$\begin{bmatrix} 2 & 5 & 1 \\ 4 & 2 & 3 \end{bmatrix} \bullet \begin{bmatrix} 2 & 5 \\ 1 & 1 \\ 3 & 2 \end{bmatrix}$$

$p \times q$ $\qquad$ $q \times r$

$$= \begin{bmatrix} 2 \bullet 2 + 5 \bullet 1 + 1 \bullet 3 & 4 \bullet 2 + 2 \bullet 1 + 3 \bullet 3 \\ 2 \bullet 5 + 5 \bullet 1 + 1 \bullet 2 & 4 \bullet 5 + 2 \bullet 1 + 3 \bullet 2 \end{bmatrix}$$

$p \times r$

Cost: Number of scalar multiplications = **pqr**

Example

Matrix	Dimensions
A_1	13 x 5
A_2	5 X 89
A_3	89 X 3
A_4	3 X 34

	Parenthesization	Scalar multiplications
1	$((A_1\ A_2)\ A_3)\ A_4$	10,582
2	$(A_1\ A_2)\ (A_3\ A_4)$	54,201
3	$(A_1\ (A_2\ A_3))\ A_4$	2, 856
4	$A_1\ ((A_2\ A_3)\ A_4)$	4, 055
5	$A_1\ (A_2\ (A_3\ A_4))$	26,418

13 x 5 x 89 *scalar multiplications* to get (A_1A_2)

13 x 89 x 3 *scalar multiplications* to get $((A_1A_2)A_3)$

13 x 3 x 34 *scalar multiplications* to get $(((A_1A_2)A_3)\ A_4)$

Dynamic Programming Approach

- The structure of an optimal solution
  - Let us use the notation $A_{i..j}$ for the matrix that results from the product $A_i A_{i+1} \ldots A_j$
  - An optimal parenthesization of the product $A_1A_2\ldots A_n$ splits the product between A_k and A_{k+1} for some integer k where$1 \leq k < n$
  - First compute matrices $A_{1..k}$ and $A_{k+1..n}$; then multiply them to get the final matrix $A_{1..n}$

Dynamic Programming Approach

…contd

- **Key observation**: parenthesizations of the subchains $A_1A_2\ldots A_k$ and $A_{k+1}A_{k+2}\ldots A_n$ must also be optimal if the parenthesization of the chain $A_1A_2\ldots A_n$ is optimal.

- That is, the optimal solution to the problem contains within it the optimal solution to subproblems.

Dynamic Programming Approach

…contd

- Recursive definition of the value of an optimal solution.
  - Let *m[i, j]* be the minimum number of scalar multiplications necessary to compute $A_{i..j}$
  - Minimum cost to compute $A_{1..n}$ is m[1, n]
  - Suppose the optimal parenthesization of $A_{i..j}$ splits the product between A_k and A_{k+1} for some integer k where $i \leq k < j$

Dynamic Programming Approach

…contd

- $A_{i..j} = (A_i A_{i+1} \ldots A_k) \cdot (A_{k+1} A_{k+2} \ldots A_j) = A_{i..k} \cdot A_{k+1..j}$
- Cost of computing $A_{i..j}$ = *cost of computing* $A_{i..k}$ + *cost of computing* $A_{k+1..j}$ + *cost of multiplying* $A_{i..k}$ *and* $A_{k+1..j}$
- Cost of multiplying $A_{i..k}$ and $A_{k+1..j}$ is $p_{i-1} p_k p_j$
- $m[i, j] = m[i, k] + m[k+1, j] + p_{i-1} p_k p_j$ *for* $i \leq k < j$
- $m[i, i] = 0$ for $i = 1, 2, \ldots, n$

Dynamic Programming Approach

…contd

- But… *optimal* parenthesization occurs at one value of k among all possible $i \le k < j$
- *Check* all these and select the *best* one

$$m[i, j] = \begin{cases} 0 & \text{if } i=j \\ \min\limits_{i \le k < j} \{m[i, k] + m[k+1, j] + p_{i-1} p_k p_j\} & \text{if } i<j \end{cases}$$

Dynamic Programming Approach

…contd

- To keep track of how to construct an optimal solution, we use a *table s*
- *s[i, j]* = value of *k* at which $A_i A_{i+1} \ldots A_j$ is split for *optimal* parenthesization.

Ex:-

$[A_1]_{5\times4}$ $[A_2]_{4\times6}$ $[A_3]_{6\times2}$ $[A_4]_{2\times7}$

$P_0=5$, $p_1=4$, $p_2=6$, $p_3=2$, $p_4=7$

Sequence Size	1	2	3	4
1	$M_{11}=0$	$M_{22}=0$	$M_{33}=0$	$M_{44}=0$
2	$M_{12}=120$ K=1	$M_{23}=48$ K=2	$M_{34}=84$ K=3	
3	$M_{13}=88$ K=1	$M_{24}=104$ K=2		
4	$M_{14}=158$ K=3			

Constructing Optimal Solution

- The *optimal solution* can be constructed from the above table.
  - Each entry *k value tells us* where to split the product $A_i A_{i+1} \ldots A_j$ for the minimum cost.

$$(A_1 \ (A_2 \ A_3) \ ) \ A_4$$

Optimal Binary Search Tree(OBST)

- *A binary search tree T is a binary tree, either it is empty or each node in the tree contains an identifier and,*
  - All identifiers in the left subtree of T are *less than* the identifier in the *root* node T.
  - All identifiers in the right subtree are *greater than* the identifier in the *root* node T.
  - The *left and right* subtree of T are also *binary search* trees.

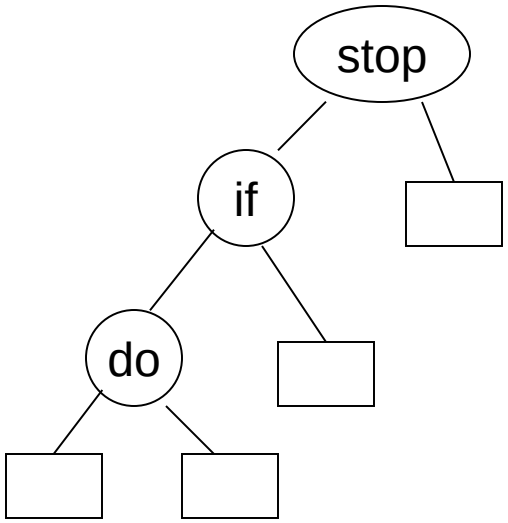
- Ex:- (a1,a2,a3)=(do,if,stop)

Here n=3

- The number of possible binary search trees= $(1/n+1)2n_{cn}$

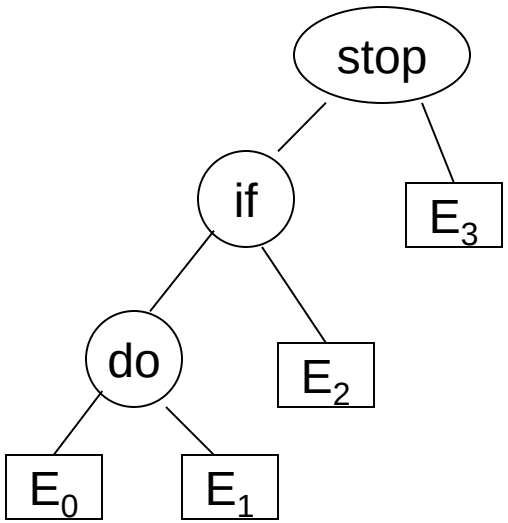
$= ¼(6c_3)$

$=5$



Optimal Binary Search Trees

- Problem
  - Given sequence of identifiers (a_1,a_2 …., $a_{n)}$ with $a_1 < a_2 < \cdots < a_n$.
  - Let *p(i) be the* probability *with which we search for* a_i
  - Let q(i) be the probability with which we search for an identifier x such that $a_i < x < a_{i+1}$.
  - Want to build a binary search tree (BST) with minimum expected search cost.



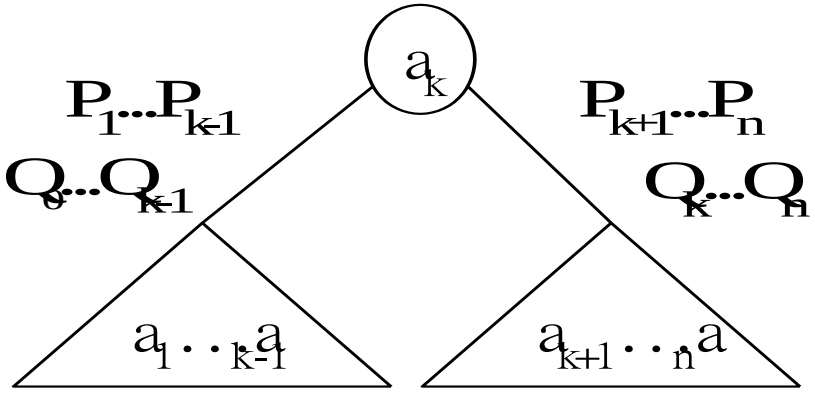
- Identifiers : stop, if, do
  Internal node : successful search, p(i)
- External node : unsuccessful search, q(i)

- The expected cost of a binary tree:

$$\sum_{1 \le i \le n} P_i * \text{level}(a_i) + \sum_{0 \le i \le n} Q_i * (\text{level}(E_i) - 1)$$

The dynamic programming approach

- Make a decision as which of the a_i's should be assigned to the root node of the tree.
- If we choose a_k, then it is clear that the internal nodes for $a_1, a_2, \ldots\ldots, a_{k-1}$ as well as the external nodes for the classes E_0, $E_1, \ldots, E_{k-1}$ will lie in the left subtree *l* of the root. The remaining nodes will be in the right subtree r.



$$\text{cost}(l) = \sum_{1 \le i < k} p(i) * \text{level}(a_i) + \sum_{0 \le i < k} q(i) * (\text{level}(E_i) - 1)$$

$$\text{cost}(r) = \sum_{k < i \le n} p(i) * \text{level}(a_i) + \sum_{k < i \le n} q(i) * (\text{level}(E_i) - 1)$$

- In both the cases the level is measured by considering the root of the respective subtree to be at level 1.

- Using $w(i, j)$ to represent the sum $q(i) + \sum_{l=i+1}^{j} (q(l) + p(l))$, we obtain the following as the expected cost of the above search tree.

$$p(k) + \text{cost}(l) + \text{cost}(r) + w(0, k-1) + w(k, n)$$

- If we use c(i,j) to represent the cost of an optimal binary search tree t_{ij} containing $a_{i+1},\ldots\ldots,a_j$ and $E_i,\ldots,E_j$, then cost(l)=c(0,k-1), and cost(r)=c(k,n).

- For the tree to be optimal, we must choose k such that

p(k) + c(0,k-1) + c(k,n) + w(0,k-1) + w(k,n) is minimum.

Hence, for c(0,n) we obtain

$$c(0,n)= \min_{1 \le k \le n} \left\{ c(0,k-1) + c(k, n) + p(k) + w(0,k-1) + w(k,n) \right\}$$

We can generalize the above formula for any c(i,j) as shown below

$$c(i, j)= \min_{i< k \le j} \left\{ c(i,k-1) + c(k,j) + \underbrace{p(k)+ w(i,k-1) + w(k,j)} \right\}$$

$$c(i, j) = \min_{i<k\leq j} \{ cost(i,k-1) + cost(k,j) \} + w(i, j)$$

- Therefore, c(0,n) can be solved by first computing all c(i, j) such that j - i=1, next we compute all c(i,j) such that j - i=2, then all c(i, j) with j – i=3, and so on.

- During this computation we record the root r(i, j) of each tree t_{ij}, then an optimal binary search tree can be constructed from these r(i, j).

- r(i, j) is the value of k that minimizes the cost value.

Note:1. c(i,i) = 0, w(i, i) = q(i), and r(i, i) = 0 for all $0 \leq i \leq n$

2. w(i, j) = p(j) + q(j) + w(i, j-1)

Ex 1: Let n=4, and (a1,a2,a3,a4) = (do, if, int, while).

Let p(1 : 4) = (3, 3, 1, 1) and q(0: 4) = (2, 3, 1,1,1).

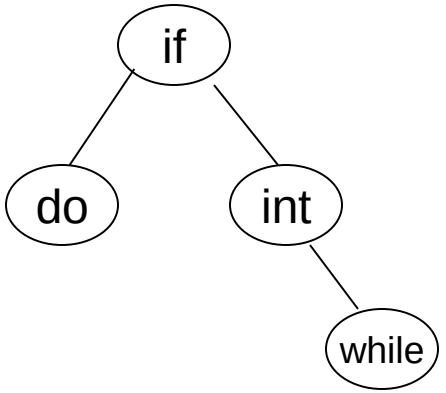
p’s and q’s have been multiplied by 16 for convenience.

Then, we get

j-i					
0	$w_{00}=2$ $c_{00}=0$ $r_{00}=0$	$w_{11}=3$ $c_{11}=0$ $r_{11}=0$	$w_{22}=1$ $c_{22}=0$ $r_{22}=0$	$w_{33}=1$ $c_{33}=0$ $r_{33}=0$	$w_{44}=1$ $c_{44}=0$ $r_{44}=0$
1	$w_{01}=8$ $c_{01}=8$ $r_{01}=1$	$w_{12}=7$ $c_{12}=7$ $r_{12}=2$	$w_{23}=3$ $c_{23}=3$ $r_{23}=3$	$w_{34}=3$ $c_{34}=3$ $r_{34}=4$	
2	$w_{02}=12$ $c_{02}=19$ $r_{02}=1$	$w_{13}=9$ $c_{13}=12$ $r_{13}=2$	$w_{24}=5$ $c_{24}=8$ $r_{24}=3$		
3	$w_{03}=14$ $c_{03}=25$ $r_{03}=2$	$w_{14}=11$ $c_{14}=19$ $r_{14}=2$			
4	$w_{04}=16$ $c_{04}=32$ $r_{04}=2$				

Computation of c(0,4), w(0,4), and r(0,4)

- From the table we can see that *c(0,4)=32* is the minimum cost of a binary search tree for *(a1, a2, a3, a4).*
- The root of tree t_{04} is $a_{2.}$
- The left subtree is t_{01} and the right subtree t_{24}.
- Tree t_{01} has root a_1; its left subtree is t_{00} and right subtree t_{11}.
- Tree t_{24} has root a_3; its left subtree is t_{22} and right subtree t_{34}.
- Thus we can construct *OBST*.



Ex 2: Let n=4, and (a1,a2,a3,a4) = (count, float, int,while).

Let p(1 : 4) =(1/20, 1/5, 1/10, 1/20) and

q(0: 4) = (1/5,1/10, 1/5,1/20,1/20).

Using the r(i, j)'s construct an optimal binary search tree.

Note : If you get a problem like this (fractions), multiply every value with the LCM to get the integers, then do the problem.

Here LCM of 5, 10, and 20 is 20. Now multiply p and q values with 20.

p = (1, 4, 2, 1)

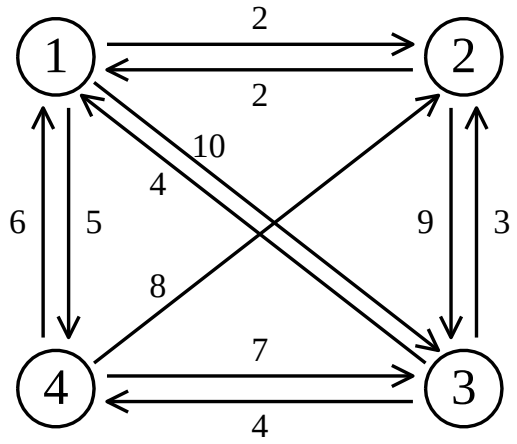
q = (4, 2, 4, 1, 1)

Traveling Salesperson Problem (TSP)

Problem:-

- You are given a set of *n cities.*
- You are given the *distances* between the cities.
- You *start and terminate* your tour at your *home city.*
- You must visit each other city *exactly once*.
- Your mission is to *determine* the *shortest tour*. OR *minimize* the *total distance* traveled.

- e.g. a directed graph :



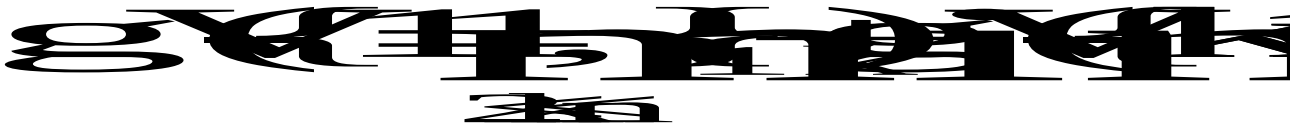
- Cost matrix:

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 2 & 10 & 5 \\ 2 & 0 & 9 & \infty \\ 4 & 3 & 0 & 4 \\ 6 & 8 & 7 & 0 \end{bmatrix} \end{array}$$

The dynamic programming approach

- Let *g(i, S)* be the length of a *shortest* path starting at vertex *i*, going through all vertices in *S* and terminating at vertex 1.

- The *length* of an optimal tour :



→ 1

- The general form:

→ 2

- Equation 1 can be solved for *g(1, V- {1})* if we know *g(k, V- {1,k})* for all choices of *k*.

- The *g* values can be obtained by using *equation 2* .

Clearly,

$$g(i, \emptyset) = C_{i1} , \quad 1 \leq i \leq n.$$

- Hence we can use *eq 2* to obtain *g(i, S)* for all *S* of *size 1*. Then we can obtain *g(i, s)* for all *S* of *size 2* and so on.

Thus,

$g(2, \emptyset)=C_{21}=2$, $g(3, \emptyset)=C_{31}=4$

$g(4, \emptyset)=C_{41}=6$

We can obtain

$g(2, \{3\})=C_{23} + g(3, \emptyset)=9+4=13$

$g(2, \{4\})=C24 + g(4, \emptyset)=\infty$

$g(3, \{2\})=C32 + g(2, \emptyset)=3+2=5$

$g(3, \{4\})=C34 + g(4, \emptyset)=4+6=10$

$g(4, \{2\}) = C42 + g(2, \emptyset) = 8+2=10$

$g(4, \{3\}) = C43 + g(3, \emptyset) = 7+4=11$

Next, we compute $g(i,S)$ with $|S| = 2$,

$g(2,\{3,4\}) = \min \{ c23+g(3,\{4\}), c24+g(4,\{3\}) \}$

$= \min \{19, \infty\} = 19$

$g(3,\{2,4\}) = \min \{ c32+g(2,\{4\}), c34+g(4,\{2\}) \}$

$= \min \{\infty, 14\} = 14$

$g(4,\{2,3\}) = \min \{c42+g(2,\{3\}), c43+g(3,\{2\}) \}$

$= \min \{21,12\} = 12$

Finally,

We obtain

$$g(1,\{2,3,4\}) = \min\{\ c12 + g(2,\{3,4\}),\ c13 + g(3,\{2,4\}),\ c14 + g(4,\{2,3\})\ \}$$

$$= \min\{2+19, 10+14, 5+12\}$$

$$= \min\{21, 24, 17\}$$

$$= 17.$$

- A tour can be constructed if we retain with each $g(i, s)$ the value of j that *minimizes* the tour distance.
- Let $J(i, s)$ be this value, then $J(1, \{2, 3, 4\}) = 4$.
- Thus the tour starts from *1* and goes to *4.*

- The remaining tour can be obtained from $g(4, \{2,3\})$. So $J(4, \{3, 2\})=3$

- Thus the next edge is $\langle 4, 3 \rangle$. The remaining tour is $g(3, \{2\})$. So $J(3,\{2\})=2$

The optimal tour is: (1, 4, 3, 2, 1)

Tour distance is 5+7+3+2 = 17

All pairs shortest path problem

Floyd-Warshall Algorithm

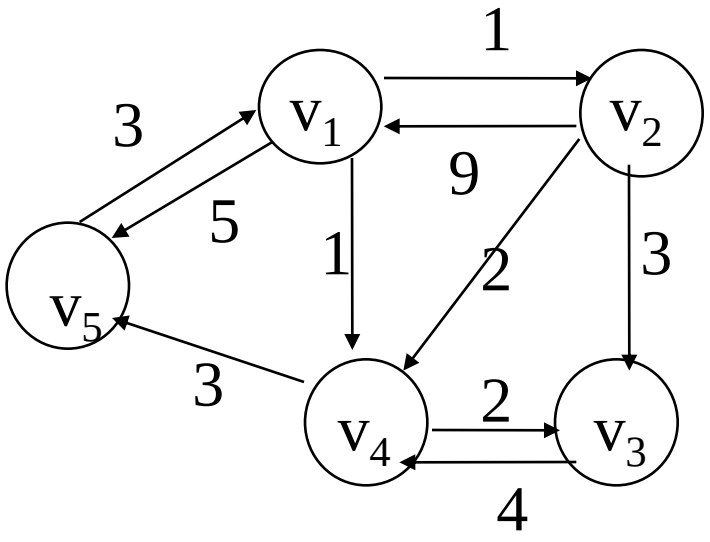
All-Pairs Shortest Path Problem

- Let G=(V,E) be a *directed* graph consisting of n vertices.
- *Weight* is associated with each edge.

- The problem is to *find a shortest path* between *every pair of nodes.*

Ex:-

	1	2	3	4	5
1	0	1	∞	1	5
2	9	0	3	2	∞
3	∞	∞	0	4	∞
4	∞	∞	2	0	3
5	3	∞	∞	∞	0



Idea of Floyd-Warshall Algorithm

- Assume vertices are *{1,2,……n}*
- Let $d^k(i, j)$ be the length of a shortest path from i to j with intermediate vertices numbered not higher than *k* where $0 \le k \le n$, then
- $d^0(i, j) = c(i, j)$ (no intermediate vertices at all)
- $d^k(i, j) = \min \{ d^{k-1}(i, j),\ d^{k-1}(i, k) + d^{k-1}(k, j) \}$
- $d^n(i, j)$ is the *length* of a shortest path from *i to j*

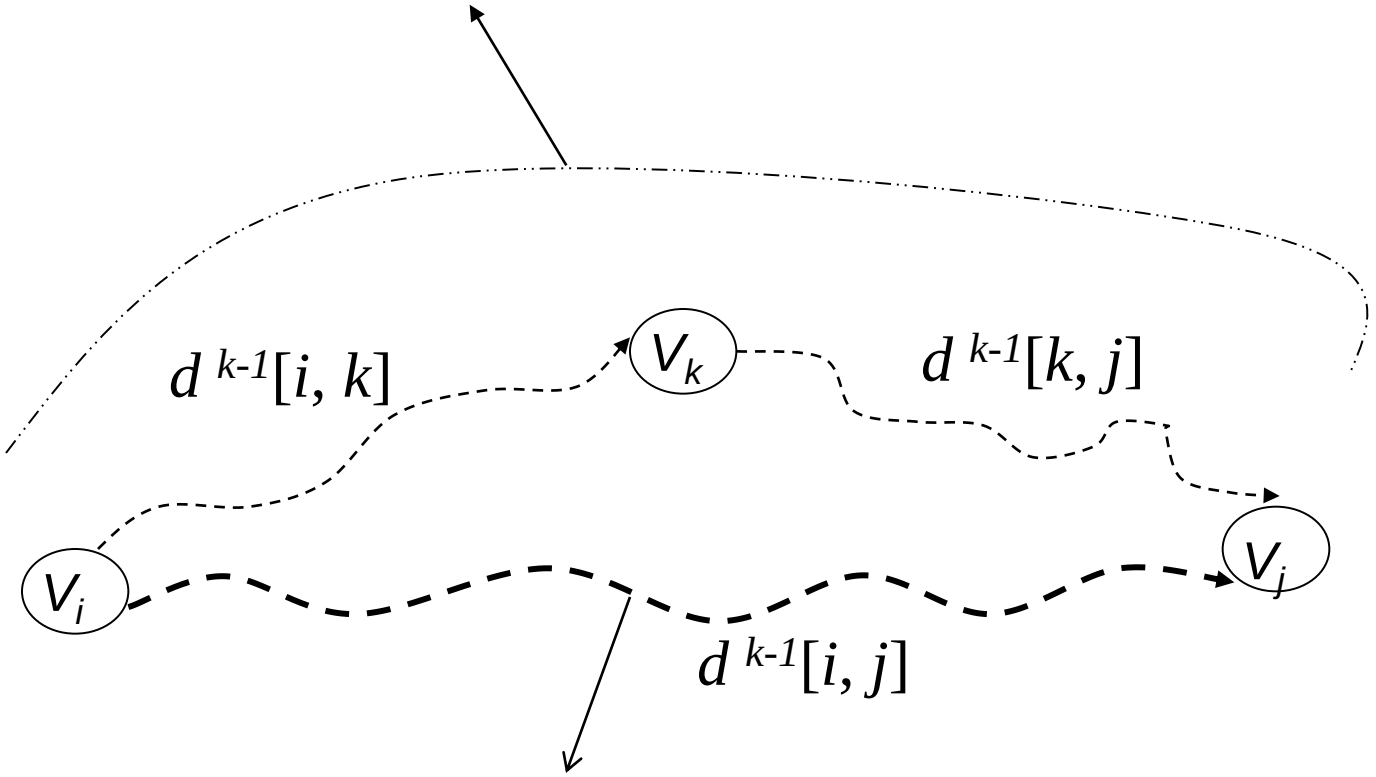
- In summary, we need to find d^{n} with d^{0} =cost matrix .

- General formula

$$d^{k}[i, j] = \min \{ d^{k-1}[i, j], d^{k-1}[i, k] + d^{k-1}[k, j] \}$$

Shortest path using intermediate vertices

$\{ V_1, \ldots V_k \}$



Shortest Path using intermediate vertices

$\{ V_{1, \ldots} V_{k-1} \}$

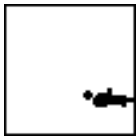
Algorithm

```
Algorithm AllPaths( c, d, n )
// c[1:n,1:n] cost matrix
// d[i,j] is the length of a shortest path from i to j
{
        for i := 1 to n do
            for j := 1 to n do
                    d [ i, j ] := c [ i, j ] ;     // copy c into d

        for k := 1 to n do
            for i := 1 to n do
                    for j := 1 to n do
                            d [ i, j ] := min ( d [ i, j ] , d [ i, k ] + d [ k, j ] );
}
```

Time Complexity is $O (n^3)$

0/1 Knapsack Problem



Let $x_i = 1$ when item i is selected and let $x_i = 0$ when item i is not selected.

$$\text{maximize} \quad \sum_{i=1}^{n} p_i x_i$$

$$\text{subject to} \quad \sum_{i=1}^{n} w_i x_i <= c$$

and $x_i = 0$ or 1 for all i

All profits and weights are positive.

- We use set s^i is a pair (P, W)
- Note That s^0 =(0, 0)
- We can compute s^{i+1} from s^i by first computing

$$S_1^i = S^i + (p_{i+1}, w_{i+1})$$

Merging :- s^{i+1} can be computed by merging the pairs in s^i and s_1^i

Purging :- if s^{i+1} contains two pairs (p_j, w_j) and (p_k, w_k) with the property that $p_j \le p_k$ and $w_j \ge w_k$ then the pair (p_j, w_j) can be discarded.

- When generating s^i's, we can also purge all pairs *(p, w)* with *w > m* as these pairs determine the value of $f_n(x)$ only for *x > m.*
- The optimal solution $f_n(m)$ is given by the *highest profit pair (last pair in* s^n *)* .

Set of 0/1 values for the x_i's

- *Set of 0/1 values for x_i's can be determined by a search through the s^i s*
  - Let (p, w) be the highest profit tuple in s^n

Step1: if $(p, w) \in s^n$ *and* $(p, w) \notin s^{n-1}$

$x_n = 1$

otherwise $x_n = 0$

This leaves us to determine how either (p, w) or $(p - p_n, w - w_n)$ *was obtained in* S^{n-1} .

This can be done recursively (Repeat Step1).

Ex: knapsack instance n=3, $(w_1, w_2, w_3)=(2,3,4)$, $(p_1,p_2,p_3)=(1,2,5)$, and m=6. for this data we have

S^0 ={ (0,0) } ; S_1^0 ={ (1,2) }

S^1 ={ (0,0), (1,2) } ; S_1^1={ (2,3), (3,5) }

S^2 ={ (0,0), (1,2), (2,3), (3,5) } ; S_1^2={ (5,4), (6,6), (7,7), (8,9) }

S^3 ={ (0,0), (1,2), (2,3), (5,4), (6,6) }

Note that the pair (3, 5) has been eliminated from s^3 as a result of the purging rule.

Also, note that the pairs (7, 7) and (8, 8) have been eliminated from s^3. Because, for these pairs w>m i.e., 7>6 and 9>6.

Therefore, the optimal solution is (6,6) (highest profit pair).

If (p, w) is the highest profit pair in sn, a set of 0/1 values for the x_i's can be determined by a search through the s^i 's.

We can set x_n=0 if (p, w) € s^{n-1} else x_n=1 . This leaves us to determine how either (p, w) or $(p - p_n, w - w_n)$ was obtained in s^{n-1}. This can be done recursively.

Solution vector is (x_1, x_2, x_3)=(1, 0, 1).